

Stage 3F Physics Note: Conformal Cartoon Translation

GL-with-AI Project

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1 Scope

Stage 3F translates the diagonal reduced cartoon geometry from physical metric variables into the conformal variables used by GRChombo-style CCZ4. It is a derivation note only. No C++ source terms or variables are implemented.

2 Metric Dictionary

For gridded indices $A, B \in \{x, z\}$,

$$\gamma_{AB} = \chi^{-1} h_{AB}, \quad \gamma_{ww} = \chi^{-1} h_{ww}. \quad (1)$$

The diagonal physical metric is therefore

$$dl^2 = \chi^{-1} h_{xx} dx^2 + \chi^{-1} h_{zz} dz^2 + x^2 \chi^{-1} h_{ww} d\Omega_2^2. \quad (2)$$

Define

$$q = \gamma_{ww} = \frac{h_{ww}}{\chi}. \quad (3)$$

The physical transverse sphere radius is

$$R_{\text{sphere}} = x \sqrt{\gamma_{ww}} = x \sqrt{\frac{h_{ww}}{\chi}}. \quad (4)$$

The angular component is

$$g_{\theta\theta} = x^2 \gamma_{ww} = \frac{x^2 h_{ww}}{\chi}, \quad (5)$$

not merely h_{ww} or γ_{ww} .

3 Determinant Condition

The conformal split is not fully specified until the conformal determinant condition is imposed. For the diagonal cartoon/Cartesian-like four-spatial-dimensional metric block,

$$\det \gamma_{4D} = \gamma_{xx} \gamma_{zz} \gamma_{ww}^2, \quad (6)$$

because the two hidden directions both contribute γ_{ww} . Thus

$$\chi = (\det \gamma_{4D})^{-1/4} = (\gamma_{xx} \gamma_{zz} \gamma_{ww}^2)^{-1/4}, \quad (7)$$

and

$$h_{xx} = \chi\gamma_{xx}, \quad h_{zz} = \chi\gamma_{zz}, \quad h_{ww} = \chi\gamma_{ww}. \quad (8)$$

The reduced conformal determinant is then

$$\det h_{4\text{D}} = h_{xx}h_{zz}h_{ww}^2 = 1. \quad (9)$$

This is the cartoon/Cartesian-like determinant condition. It does not include the spherical-coordinate Jacobian factor $x^4 \sin^2 \theta$. The angular metric component remains

$$g_{\theta\theta} = x^2\gamma_{ww} = \frac{x^2 h_{ww}}{\chi}. \quad (10)$$

4 Derivative Identities

For $q = h_{ww}/\chi$,

$$\partial_A q = \chi^{-1} \partial_A h_{ww} - \chi^{-2} h_{ww} \partial_A \chi. \quad (11)$$

The second derivative is

$$\begin{aligned} \partial_A \partial_B q &= \chi^{-1} \partial_A \partial_B h_{ww} - \chi^{-2} [\partial_A h_{ww} \partial_B \chi + \partial_B h_{ww} \partial_A \chi + h_{ww} \partial_A \partial_B \chi] \\ &\quad + 2\chi^{-3} h_{ww} \partial_A \chi \partial_B \chi. \end{aligned} \quad (12)$$

Equivalently,

$$\partial_A \log q = \partial_A \log h_{ww} - \partial_A \log \chi, \quad (13)$$

and

$$\frac{\partial_A \partial_B q}{q} = \partial_A \partial_B \log q + \partial_A \log q \partial_B \log q. \quad (14)$$

5 Physical Radius

Let

$$f = x\sqrt{q} = x\sqrt{\frac{h_{ww}}{\chi}}. \quad (15)$$

Then

$$\partial_x \log f = \frac{1}{x} + \frac{1}{2} (\partial_x \log h_{ww} - \partial_x \log \chi), \quad (16)$$

$$\partial_z \log f = \frac{1}{2} (\partial_z \log h_{ww} - \partial_z \log \chi), \quad (17)$$

and

$$\partial_A f = f \partial_A \log f. \quad (18)$$

6 Extrinsic Curvature

With `GR_SPACEDIM = 4`,

$$K_{AB} = \chi^{-1} \left(A_{AB} + \frac{h_{AB}K}{4} \right), \quad K_{ww} = \chi^{-1} \left(A_{ww} + \frac{h_{ww}K}{4} \right). \quad (19)$$

The hidden component K_{ww} is the hidden Cartesian-like physical extrinsic-curvature component for each suppressed direction.

The full four-dimensional conformal tracelessness relation is

$$h^{AB}A_{AB} + \frac{2A_{ww}}{h_{ww}} = 0. \quad (20)$$

For diagonal h_{AB} ,

$$\frac{A_{xx}}{h_{xx}} + \frac{A_{zz}}{h_{zz}} + \frac{2A_{ww}}{h_{ww}} = 0, \quad (21)$$

so

$$A_{ww} = -\frac{h_{ww}}{2} \left(\frac{A_{xx}}{h_{xx}} + \frac{A_{zz}}{h_{zz}} \right). \quad (22)$$

7 Repository Checks Added After Review

The initial Stage 3F repository script checked the $q = h_{ww}/\chi$ derivative identities, physical-radius identities, and the diagonal tracelessness solve. After review, the script also checks the determinant condition, the physical-to-conformal metric round trip, reconstruction of K_{xx} , K_{zz} , and K_{ww} , and the full four-dimensional tracelessness relation with hidden multiplicity.

It also includes a guard for the dimension denominator in

$$K_{ww} = \chi^{-1} \left(A_{ww} + \frac{h_{ww}K}{4} \right). \quad (23)$$

The script reconstructs K_{ww} with a generic denominator p . The check passes for $p = 4$, but fails as expected for $p = 2$ and $p = 3$ under a positive numeric substitution. This records that the denominator is the full `GR_SPACEDIM = 4`, not `CH_SPACEDIM = 2` or ordinary $3 + 1$ intuition.

8 Limitations

This note covers only the diagonal reduced metric. Off-diagonal h_{xz} or g_{xz} , full CCZ4 RHS terms, regularity near $x = 0$, and C++ variable layout remain future work.